

TUTEDUDE DEVOPS — ASSIGNMENT 4

Git & GitHub

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Course: Tutedude DevOps

GitHub: <https://github.com/Umair-bey/tutedude-assignments>

Date: October 2026

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1. ENVIRONMENT VERIFICATION

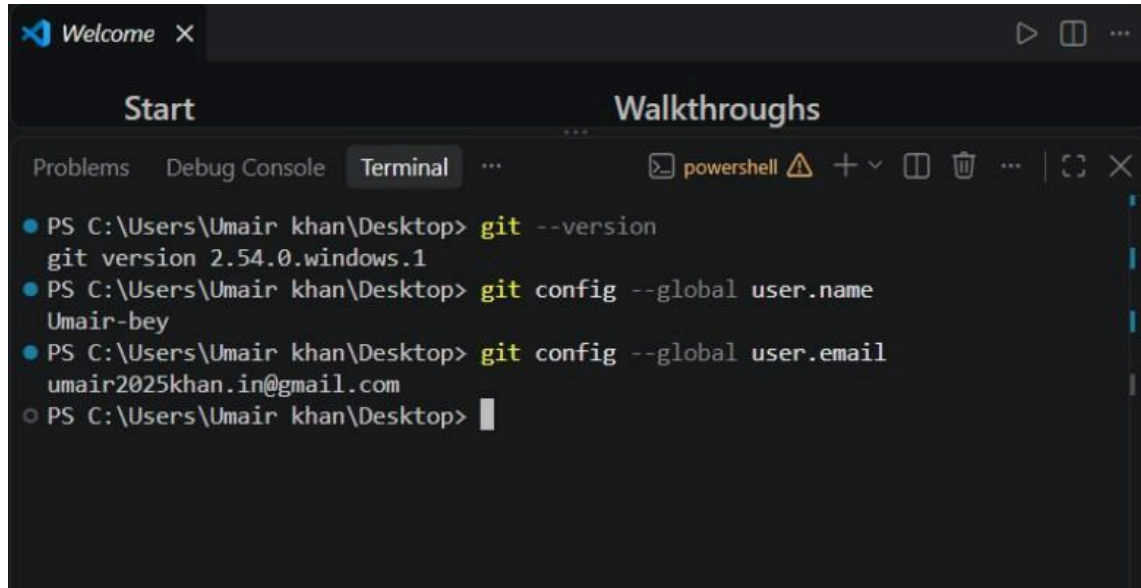
1.1 Git Version & Config

Commands:

```
git --version  
git config --global user.name  
git config --global user.email
```

Result:

```
Git version: 2.54.0.windows.1  
User: Umair-bey  
Email: umair2025khan.in@gmail.com
```



The image shows a Visual Studio Code interface with a terminal window open. The terminal is titled 'Terminal' and shows a PowerShell session. The user is in the directory C:\Users\Umair khan\Desktop. The commands and their outputs are as follows:

```
PS C:\Users\Umair khan\Desktop> git --version
git version 2.54.0.windows.1
PS C:\Users\Umair khan\Desktop> git config --global user.name
Umair-bey
PS C:\Users\Umair khan\Desktop> git config --global user.email
umair2025khan.in@gmail.com
PS C:\Users\Umair khan\Desktop>
```

1.2 Python, pip & PyMongo

Commands:

```
python --version
```

```
pip --version
```

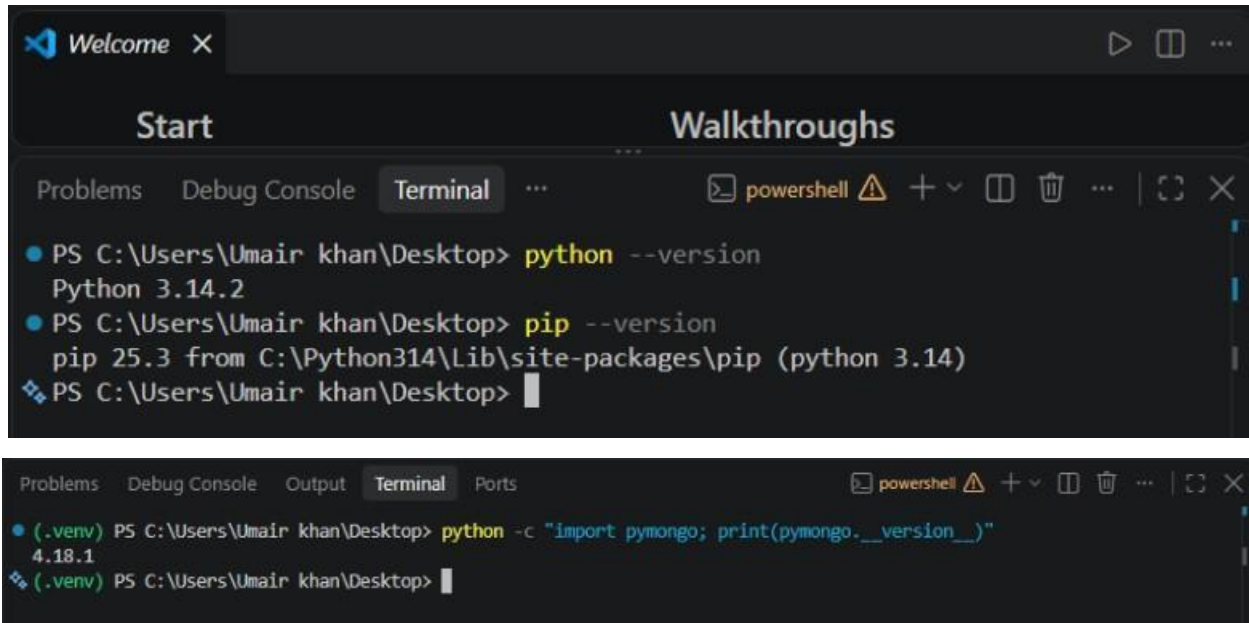
```
python -c "import pymongo; print(pymongo._version_)"
```

Result:

```
Python: 3.14.2
```

```
pip: 25.3
```

```
PyMongo: 4.18.1
```



```
PS C:\Users\Umais khan\Desktop> python --version
Python 3.14.2
PS C:\Users\Umais khan\Desktop> pip --version
pip 25.3 from C:\Python314\Lib\site-packages\pip (python 3.14)
PS C:\Users\Umais khan\Desktop>

(.venv) PS C:\Users\Umais khan\Desktop> python -c "import pymongo; print(pymongo.__version__)"
4.18.1
(.venv) PS C:\Users\Umais khan\Desktop>
```

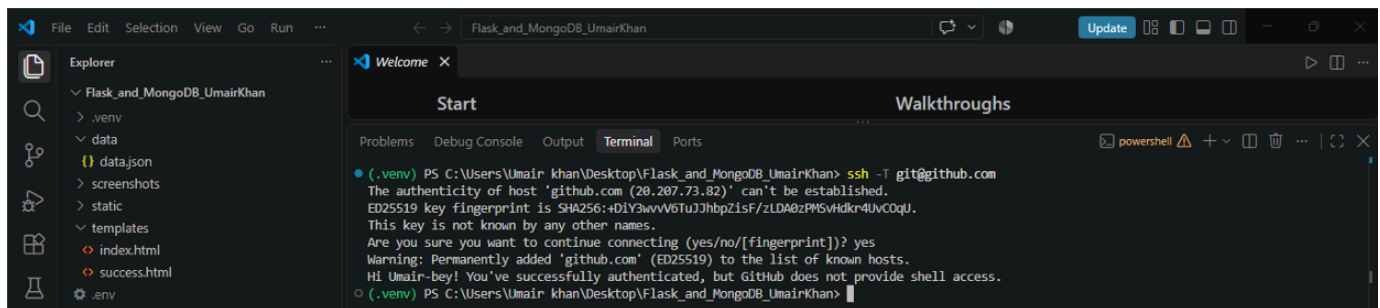
1.3 SSH Connection to GitHub

Command:

```
ssh -T git@github.com
```

Result:

Hi Umais-bey! You've successfully authenticated...



```
(.venv) PS C:\Users\Umais khan\Desktop\Flask_and_MongoDB_UmaisKhan> ssh -T git@github.com
The authenticity of host 'github.com (20.207.73.82)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wWV6TujjHbpZLisf/zLDA0zPMSvHdkr9UvCoqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi Umais-bey! You've successfully authenticated, but GitHub does not provide shell access.
(.venv) PS C:\Users\Umais khan\Desktop\Flask_and_MongoDB_UmaisKhan>
```

2. TASK 1 — REPOSITORY SETUP & FIRST BRANCH

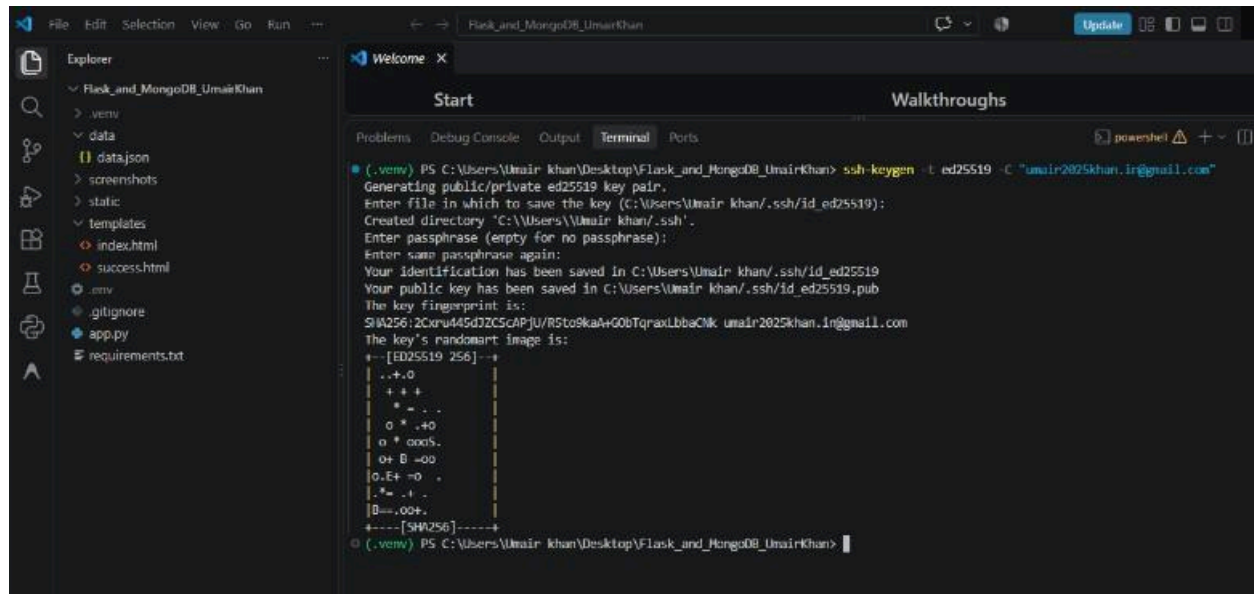
Step 1: SSH Key Generation

Command:

```
ssh-keygen -t ed25519 -C "umair2025khan.in@gmail.com"
```

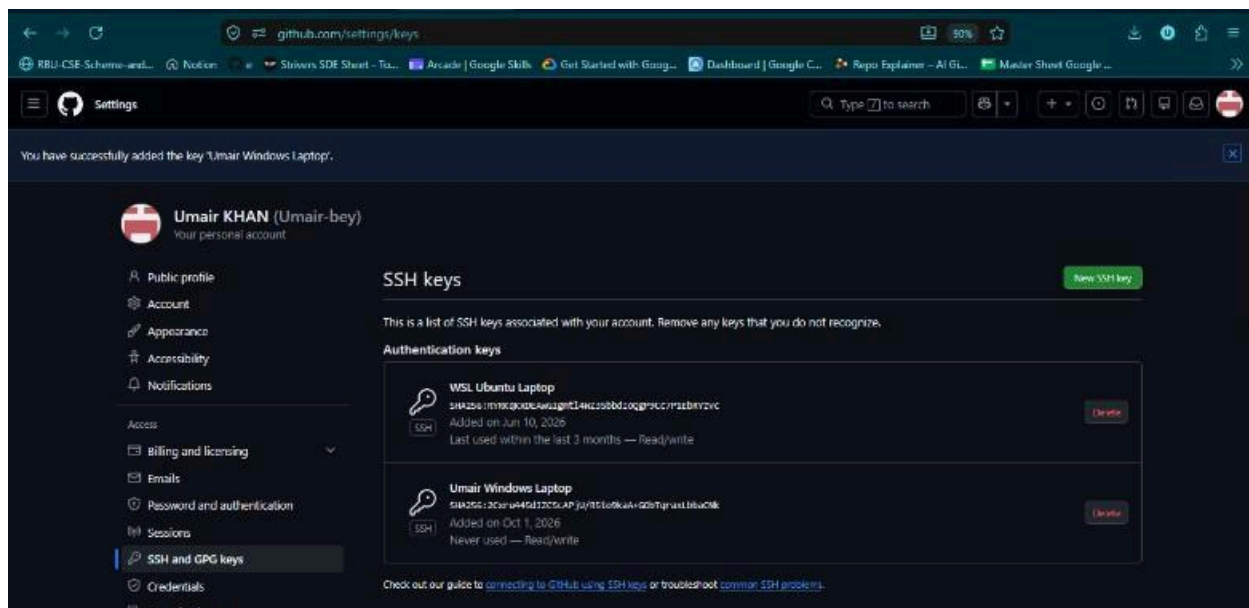
Result:

SSH key pair generated at ~/.ssh/id_ed25519 and id_ed25519.pub



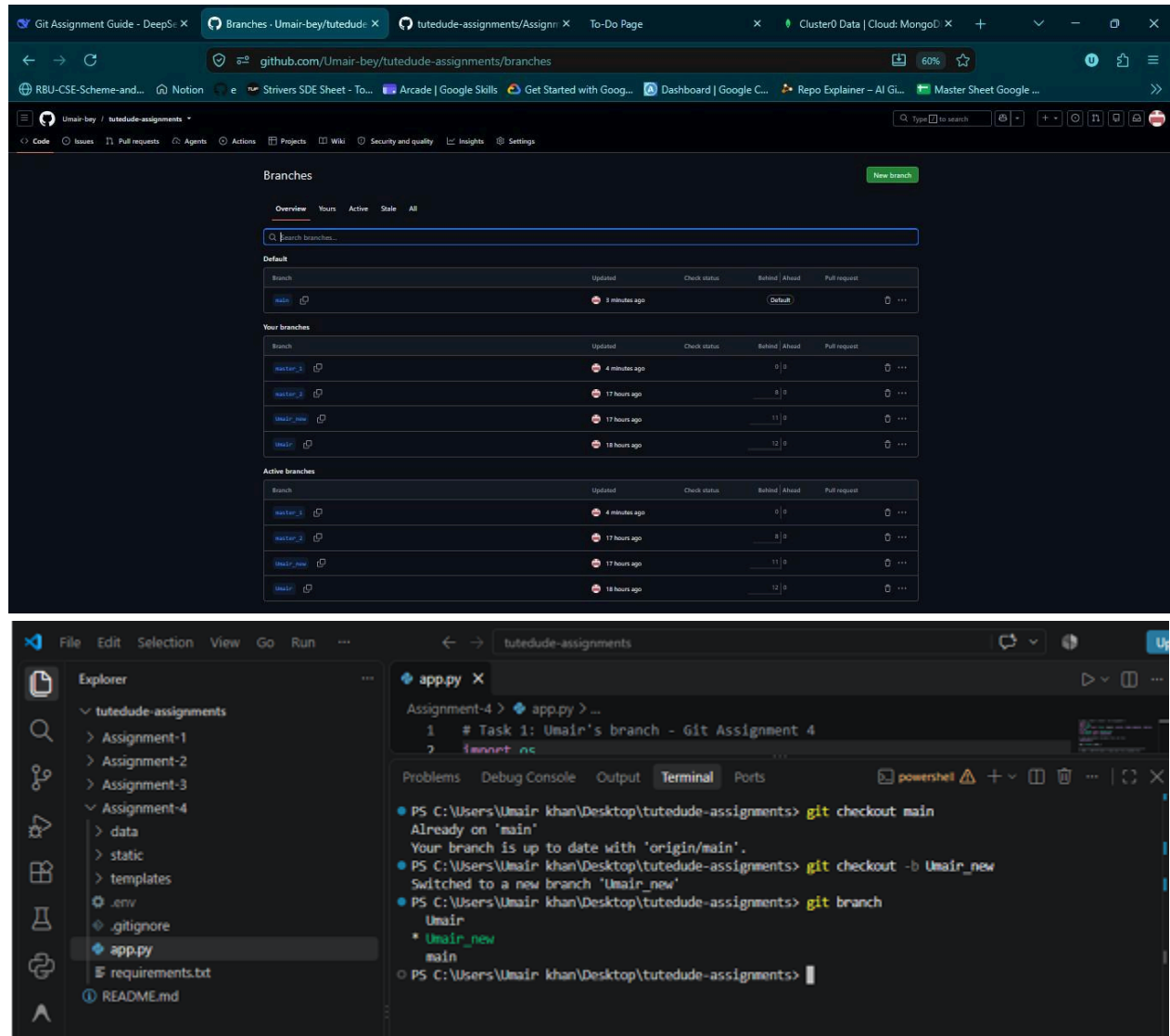
Step 2: SSH Key Added to GitHub

Key labeled "Umair Windows Laptop" added at github.com/settings/keys



Step 3: Branches Visible

Branches created: Umair, Umair_new, master_1, master_2, main



Step 4: Committed Change on Umain Branch

Commit: 600a2e1 — "Task 1: Add author comment to app.py in Umain branch"

Added author comment at the top of app.py

Step 5: Merged Umain into main

Commands:

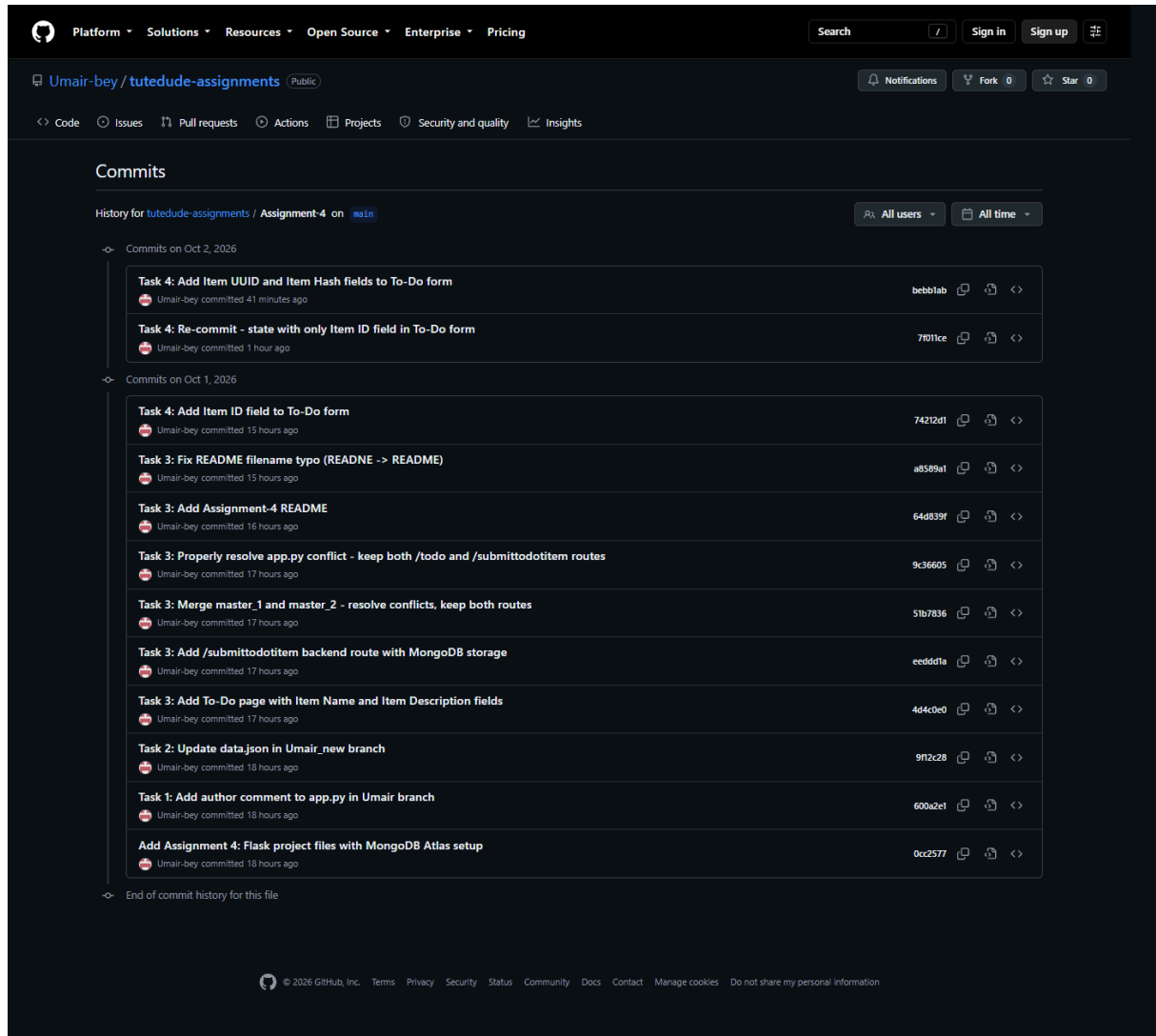
```

git checkout main
git merge Umain
git push origin main

```

Result:

Fast-forward merge completed successfully



3. TASK 2 — UPDATE JSON & RESOLVE CONFLICTS

Step 1: Created Branch Umair_new

Command:

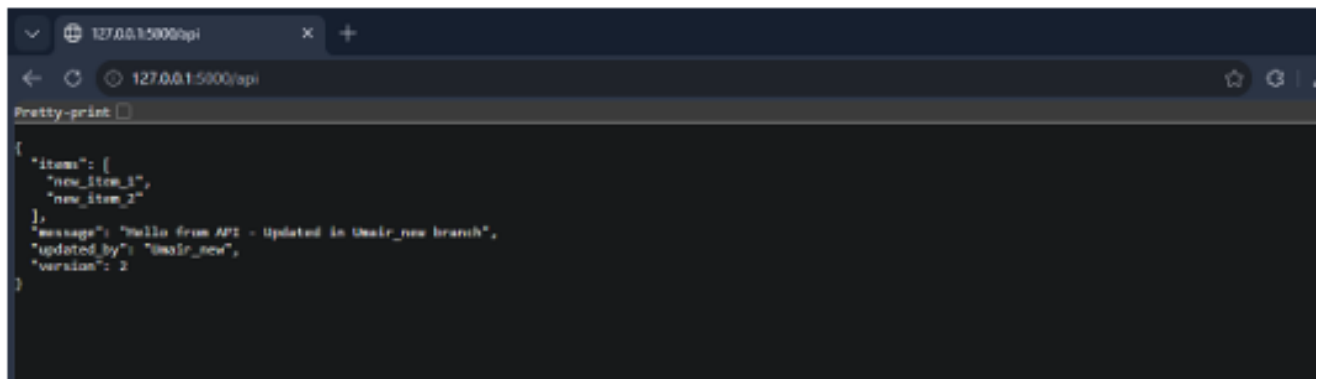
```
git checkout -b Umair_new
```

Step 2: Updated Assignment-4/data/data.json on Umair_new

New content:

```
{
  "message": "Hello from API - Updated in Umair_new branch",
  "version": 2,
  "items": ["new_item_1", "new_item_2"],
  "updated_by": "Umair_new"
}
```

Commit: 9f12c28 — "Task 2: Update data.json in Umair_new branch"



Step 3: Modified data.json on main Differently

New content:

```
{
  "message": "Hello from API - Updated in main branch",
  "version": 3,
  "items": ["main_item_1"],
  "updated_by": "main"
}
```

Commit: 2a8f742 — "Task 2: Update data.json in main branch"

Step 4: Merged Umair_new → main (CONFLICT)

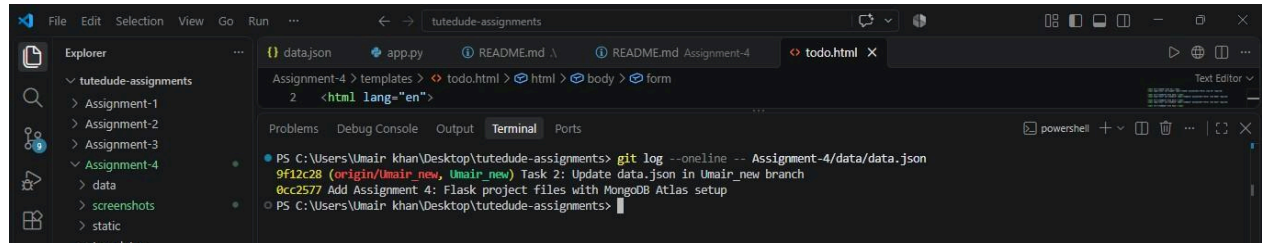
Command:

```
git merge Umair_new
```

Output:

CONFLICT (content): Merge conflict in Assignment-4/data/data.json

Automatic merge failed; fix conflicts and then commit the result.



Step 5: Conflict Markers & Resolution

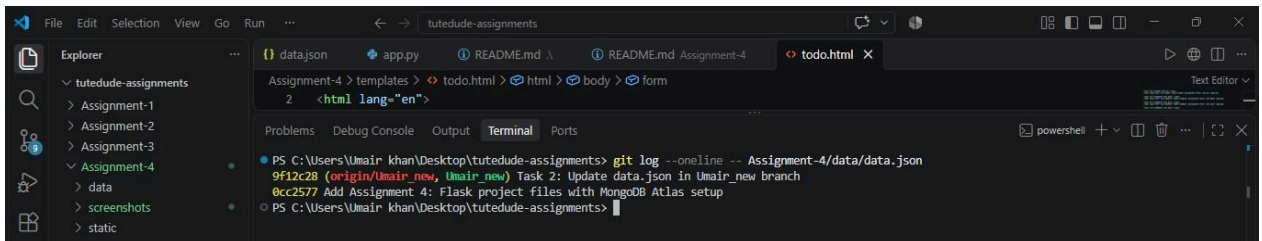
Conflict markers appeared in data.json:

```
<<<<<<< HEAD
{...main version...}
=====
{...Umair_new version...}
>>>>>>> Umair_new
```

Resolution: Accepted the Umair_new version (removed all markers)

Step 6: Staged, Committed, Pushed

Commit: 053866f — "Task 2: Resolve merge conflict - accept changes from Umair_new"



4. TASK 3 — PARALLEL FEATURE DEVELOPMENT

Step 1: Created Two Branches from main

Commands:

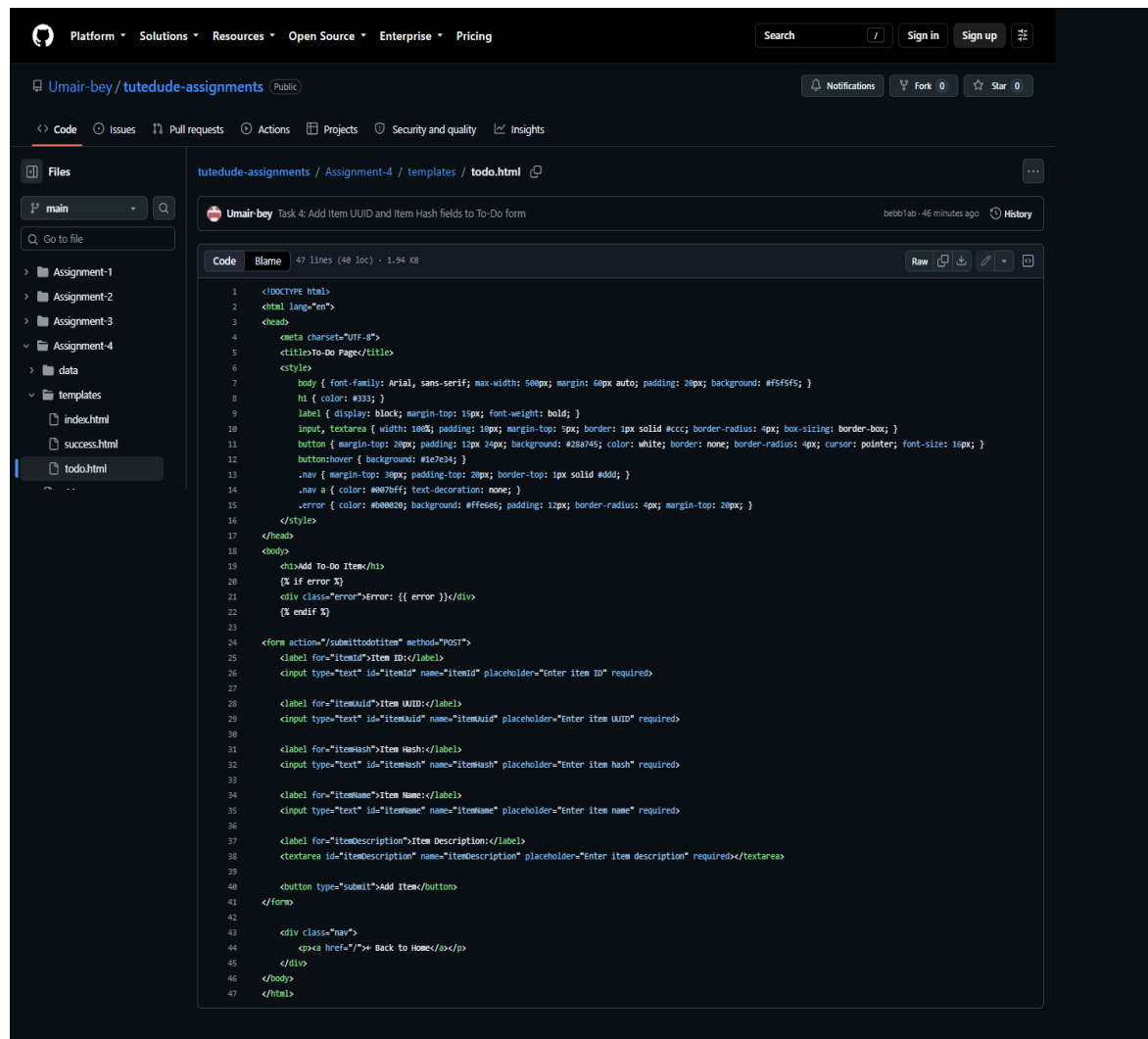
```
git checkout main
```

```
git checkout -b master_1
git checkout main
git checkout -b master_2
```

Step 2: Frontend To-Do Page in master_1

File: Assignment-4/templates/todo.html

Commit: 4d4c0e0 — "Task 3: Add To-Do page with Item Name and Item Description fields"

The screenshot shows a GitHub repository for 'Umais-bey / tutodude-assignments'. The 'Code' tab is selected, displaying the file 'Assignment-4 / templates / todo.html'. The commit history shows a commit by 'bebo1ab' 46 minutes ago. The code is a single HTML file with 47 lines, 48 loc, and 1.94 KB. It includes a form for adding a new item to a To-Do list, with fields for item ID, UUID, hash, name, and description. The form is styled with a dark theme and includes a submit button. The code is as follows:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <title>To-Do Page</title>
6   <style>
7     body { font-family: Arial, sans-serif; max-width: 500px; margin: 60px auto; padding: 20px; background: #f5f5f5; }
8     h1 { color: #333; }
9     label { display: block; margin-top: 10px; font-weight: bold; }
10    input, textarea { width: 100%; padding: 10px; margin-top: 5px; border: 1px solid #ccc; border-radius: 4px; box-sizing: border-box; }
11    button { margin-top: 20px; padding: 12px 24px; background: #2196f3; color: white; border: none; border-radius: 4px; cursor: pointer; font-size: 16px; }
12    button:hover { background: #00bcd4; }
13    .nav { margin-top: 30px; padding-top: 20px; border-top: 1px solid #ddd; }
14    .nav a { color: #007bff; text-decoration: none; }
15    .error { color: #000000; background: #ffe0e0; padding: 12px; border-radius: 4px; margin-top: 20px; }
16  </style>
17 </head>
18 <body>
19   <h1>Add To-Do Item</h1>
20   <!-- if error -->
21   <div class="error">{{ error }}</div>
22   <!-- end if -->
23
24   <form action="/submittodotitem" method="POST">
25     <label for="itemid">Item ID</label>
26     <input type="text" id="itemid" name="itemid" placeholder="Enter item ID" required>
27
28     <label for="itemuid">Item UUID</label>
29     <input type="text" id="itemuid" name="itemuid" placeholder="Enter item UUID" required>
30
31     <label for="itemhash">Item Hash</label>
32     <input type="text" id="itemhash" name="itemhash" placeholder="Enter item hash" required>
33
34     <label for="itemname">Item Name</label>
35     <input type="text" id="itemname" name="itemname" placeholder="Enter item name" required>
36
37     <label for="itemdescription">Item Description</label>
38     <textarea id="itemdescription" name="itemdescription" placeholder="Enter item description" required></textarea>
39
40     <button type="submit">Add Item</button>
41   </form>
42
43   <div class="nav">
44     <a href="/">Back to Home</a></div>
45 </body>
46 </html>
```

Step 3: Backend API in master_2

Added /submittodotitem POST route to Assignment-4/app.py

Routes visible: /api, /submit, /success, /todo, /submittodotitem

Commit: eeddd1a — "Task 3: Add /submittodotitem backend route with MongoDB storage"

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Umair-bey / tutode-assignmentsPublic

NotificationsFork 0

<> CodeIssuesPull requestsActionsProjectsSecurity and qualityInsights

Files

main

Go to file

Assignment-1

Assignment-2

Assignment-3

Assignment-4

data

templates

.gitignore

README.md

app.py

tutode-assignments / Assignment-4 / app.py

Umair-bey Task 3: Properly resolve app.py conflict - keep both /todo and /submi... 9c36605 · 17 hours ago

CodeBlame88 lines (70 loc) · 2.61 KB

RawDownloadEditView

```
1 # Task 1: Umair's branch - Git Assignment 4
2 import os
3 import json
4 from flask import Flask, request, jsonify, render_template, redirect, url_for
5 from pymongo import MongoClient
6 from dotenv import load_dotenv
7
8 # Load environment variables from .env file
9 load_dotenv()
10
11 app = Flask(__name__)
12
13 # =====
14 # TASK 1: JSON API Route - reads data from a backend file
15 # =====
16 DATA_FILE = os.path.join(os.path.dirname(__file__), 'data', 'data.json')
17
18 def read_data_file():
19     """Read the JSON list from the backend file."""
20     with open(DATA_FILE, 'r') as f:
21         return json.load(f)
22
23 @app.route('/api', methods=['GET'])
24 def api():
25     """Return the JSON list stored in data.json"""
26     data = read_data_file()
27     return jsonify(data)
28
29 # =====
30 # TASK 2: Frontend Form + MongoDB Atlas
31 # =====
32 MONGO_URI = os.getenv("MONGO_URI")
33 client = MongoClient(MONGO_URI)
34 db = client.get_database('tutode_db')
35 collection = db['submissions']
36
37
38 @app.route('/')
39 def index():
40     """Show the form."""
41     return render_template('index.html')
42
43
44 @app.route('/submit', methods=['POST'])
45 def submit():
46     """Handle form submission."""
47     try:
48         data = request.form.to_dict()
49         if not data.get('name') or not data.get('email'):
50             raise ValueError("Name and Email are required.")
51         collection.insert_one(data)
52         # Success: redirect to a different page
53         return redirect(url_for('success'))
54     except Exception as e:
55         # Error: render the same page with the error message
56         return render_template('index.html', error=str(e))
57
58
59 @app.route('/success')
60 def success():
61     """Success page shown after form submission."""
62     return render_template('success.html')
63
64
65 @app.route('/todo')
66 def todo():
67     return render_template('todo.html')
68 @app.route('/submittodoitem', methods=['POST'])
69 def submit_todo_item():
70     """Handle To-Do form submission and store in MongoDB."""
71     try:
72         item_name = request.form.get('itemName')
73         item_description = request.form.get('itemDescription')
74
75         if not item_name or not item_description:
76             raise ValueError("Item Name and Item Description are required.")
77
78         collection.insert_one({
79             'itemName': item_name,
80             'itemDescription': item_description
81         })
82
83         return redirect(url_for('todo'))
84     except Exception as e:
85         return render_template('todo.html', error=str(e))
86
87 if __name__ == '__main__':
88     app.run(debug=True)
```

Symbols

Find definitions and references for function symbols in this file by clicking a symbol or code.

Filter symbols

Step 4: Merged Both Branches into main

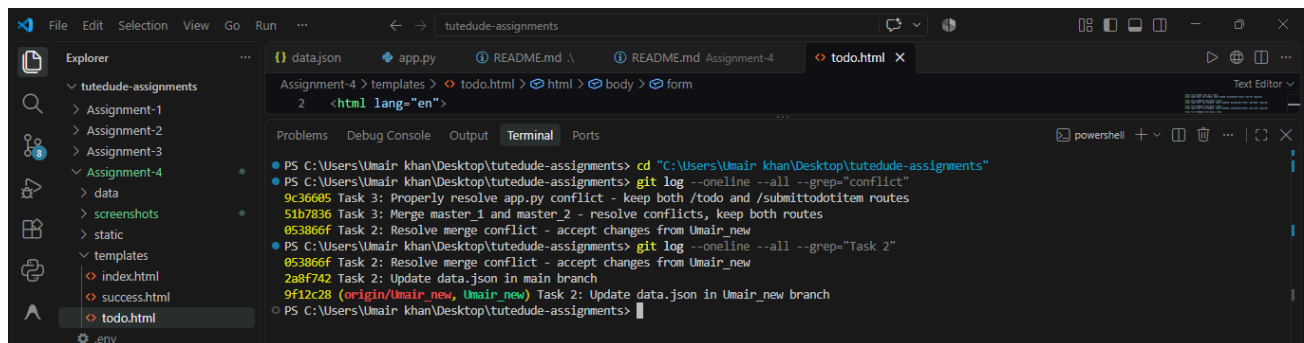
Commands:

```
git checkout main
git merge master_1
git merge master_2
```

Conflict in app.py (both branches added routes)

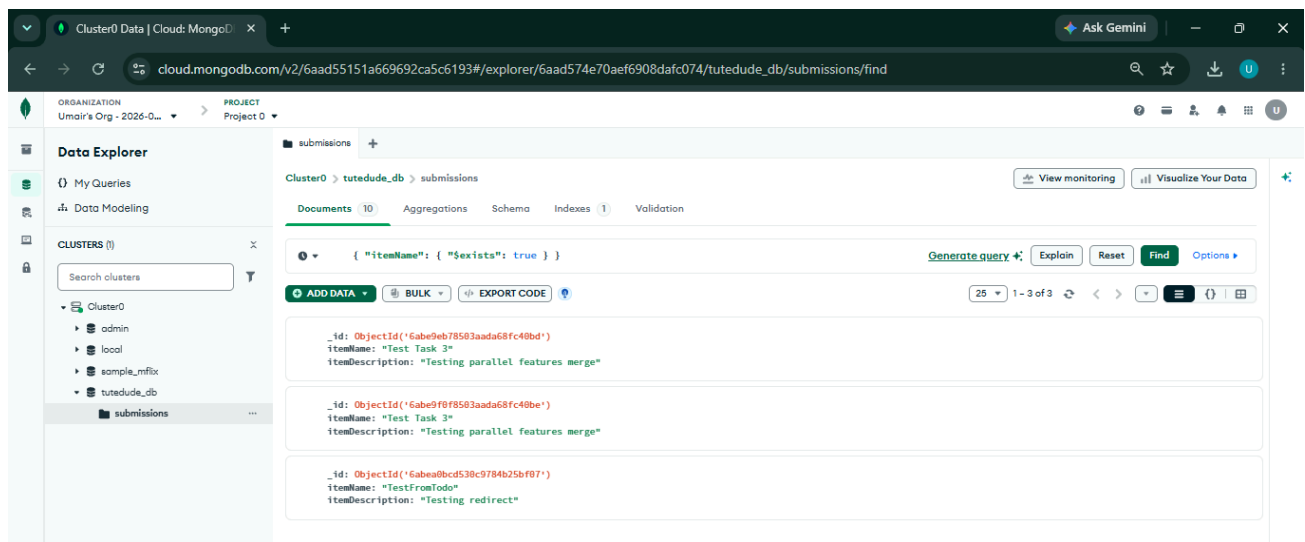
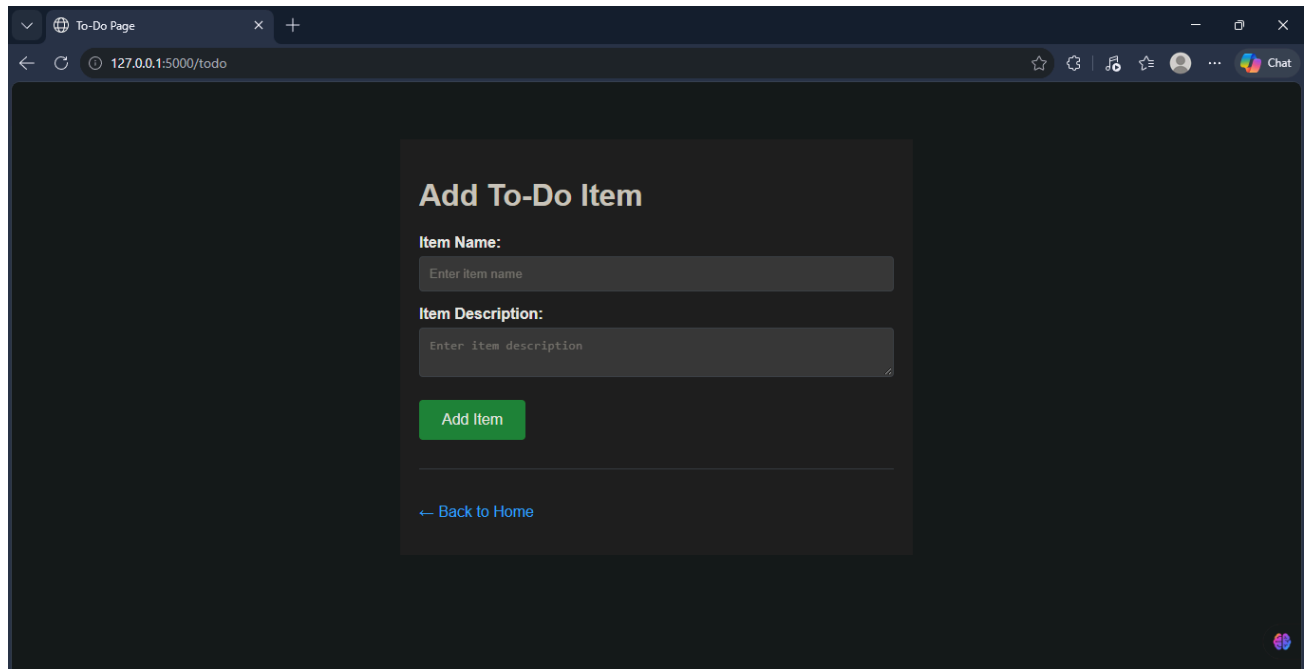
Resolution: Accepted both changes — kept /todo AND /submittodotitem

Commit: 51b7836 — "Task 3: Merge master_1 and master_2 - resolve conflicts, keep both routes"



Step 5: Tested To-Do Form End-to-End

- Flask server running on `http://localhost:5000`
- Submitted To-Do form
- Data stored in MongoDB Atlas collection `tutodude_db.submissions`



Step 6: Verified Assignment-4 Folder on GitHub

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Umais-bey / tutedude-assignments Public

Notifications Fork 0 Star 0

< Code Issues Pull requests Actions Projects Security and quality Insights

Files

main

Go to file

Assignment-1

Assignment-2

Assignment-3

Assignment-4

data

templates

.gitignore

README.md

app.py

.

tutedude-assignments / Assignment-4 /

Umais-bey Task 4: Add Item UUID and Item Hash fields to To-Do form bebb1ab · 43 minutes ago History

Name	Last commit message	Last commit date
..		
data	Task 2: Update data.json in Umais_new branch	18 hours ago
templates	Task 4: Add Item UUID and Item Hash fields to To-Do form	43 minutes ago
.gitignore	Add Assignment 4: Flask project files with MongoDB Atlas setup	18 hours ago
README.md	Task 3: Fix README filename typo (READNE -> README)	15 hours ago
app.py	Task 3: Properly resolve app.py conflict - keep both /todo and /submi...	17 hours ago
requirements.txt	Add Assignment 4: Flask project files with MongoDB Atlas setup	18 hours ago

README.md

Assignment 4: Git & GitHub

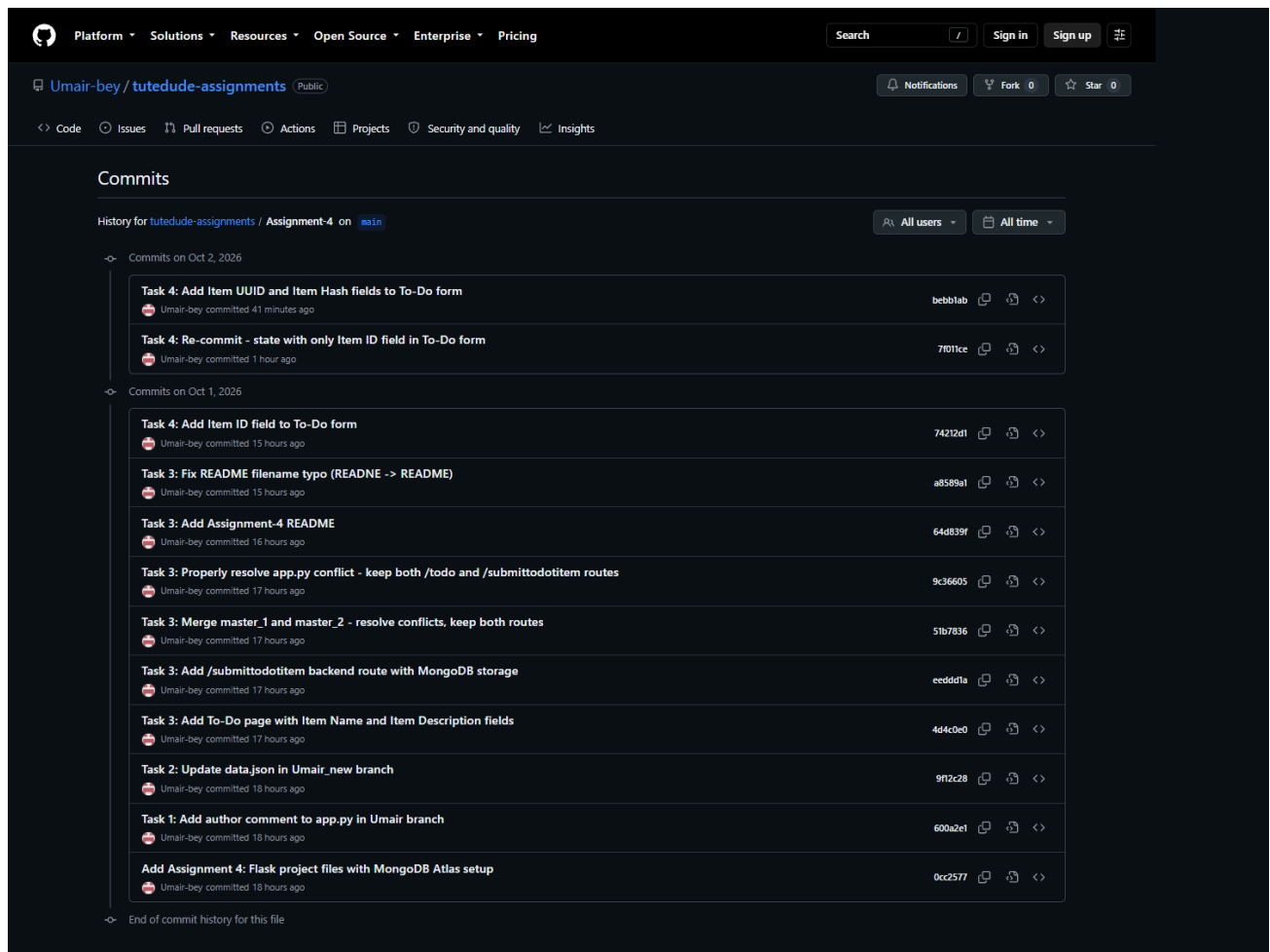
Author: Umais Khan Course: Tutedude DevOps Repo: <https://github.com/Umais-bey/tutedude-assignments>

Tasks Completed

- Task 1: Repository setup, SSH key, branch Umais
- Task 2: JSON update and merge conflict resolution
- Task 3: Parallel branches master_1 and master_2 with MongoDB Atlas
- Task 4: Sequential commits, soft reset, rebase

Routes

- / — Home form
- /api — JSON data
- /todo — To-Do form
- /submittodotitem — POST to MongoDB



=====

5. TASK 4 — SEQUENTIAL COMMITS, RESET & REBASE

=====

Step 1: Synced master_1 with main

Command:

```
git checkout master_1
```

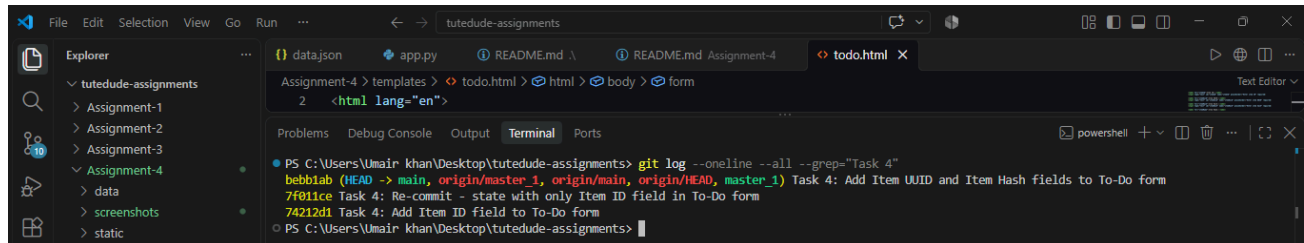
git merge main

Step 2: Added 3 Fields Sequentially (3 Separate Commits)

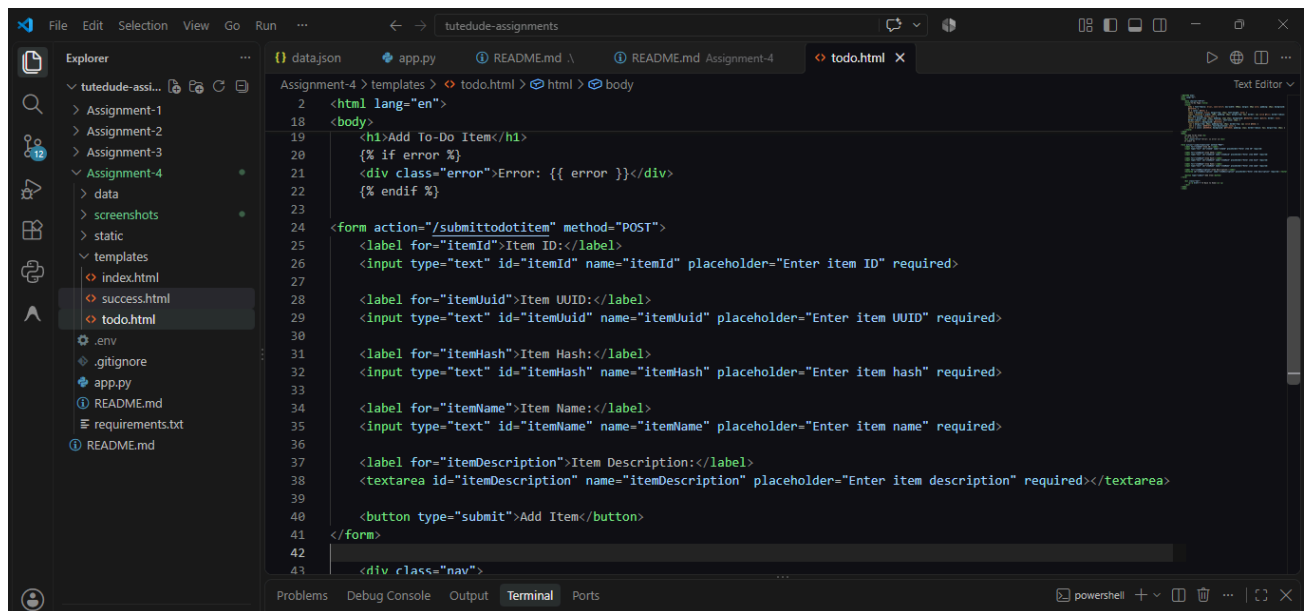
Commit 1 — 74212d1 — "Task 4: Add Item ID field to To-Do form"

Commit 2 — be02f52 — "Task 4: Add Item UUID field to To-Do form"

Commit 3 — cf2bc93 — "Task 4: Add Item Hash field to To-Do form"



```
PS C:\Users\Umair Khan\Desktop\tutodude-assignments> git log --oneline --all --grep="Task 4"
bebb1ab (HEAD -> main, origin/master 1, origin/main, origin/HEAD, master 1) Task 4: Add Item UUID and Item Hash fields to To-Do form
7f011ce Task 4: Re-commit - state with only Item ID field in To-Do form
74212d1 Task 4: Add Item ID field to To-Do form
```



```
2 <html lang="en">
18 <body>
19 <h1>Add To-Do Item</h1>
20 {% if error %}
21 <div class="error">Error: {{ error }}</div>
22 {% endif %}
23
24 <form action="/submittodotitem" method="POST">
25 <label for="itemId">Item ID:</label>
26 <input type="text" id="itemId" name="itemId" placeholder="Enter item ID" required>
27
28 <label for="itemUuid">Item UUID:</label>
29 <input type="text" id="itemUuid" name="itemUuid" placeholder="Enter item UUID" required>
30
31 <label for="itemHash">Item Hash:</label>
32 <input type="text" id="itemHash" name="itemHash" placeholder="Enter item hash" required>
33
34 <label for="itemName">Item Name:</label>
35 <input type="text" id="itemName" name="itemName" placeholder="Enter item name" required>
36
37 <label for="itemDescription">Item Description:</label>
38 <textarea id="itemDescription" name="itemDescription" placeholder="Enter item description" required></textarea>
39
40 <button type="submit">Add Item</button>
41 </form>
42
43 <div class="nav">
```

Final todo.html contains all 5 fields:

Item ID, Item UUID, Item Hash, Item Name, Item Description

Step 3: Merged master_1 into main

Command:

git checkout main

git merge master_1

Step 4: Soft Reset to Item ID Commit

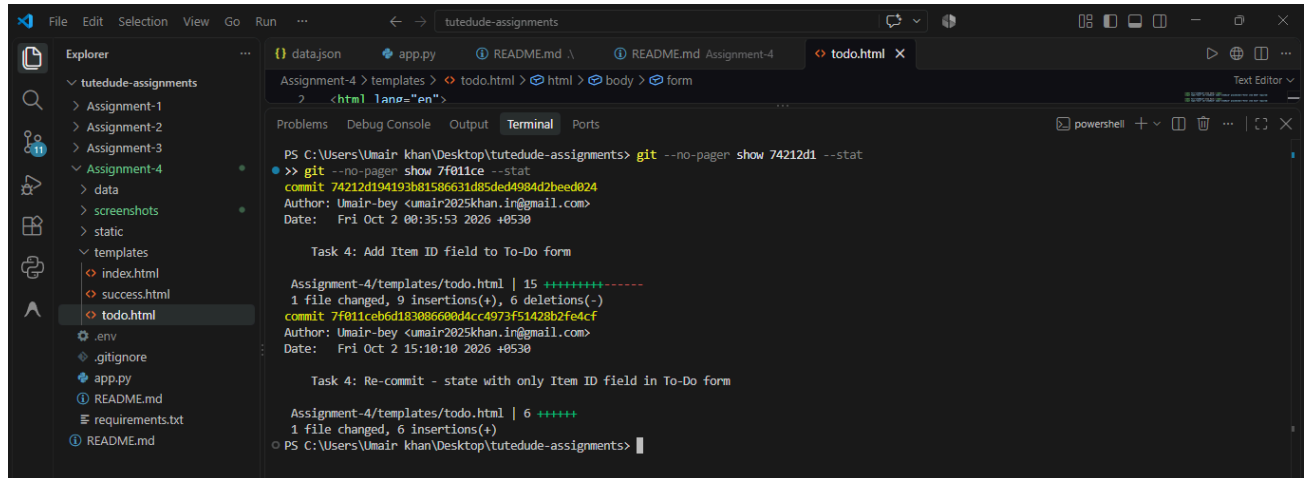
Command:

git reset --soft 74212d1

Result:

UUID and Hash changes became staged but not committed

Re-committed: 7f011ce — "Task 4: Re-commit - state with only Item ID field in To-Do form"



The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left displaying the project structure. The main editor area shows the 'todo.html' file. The bottom panel displays the 'Terminal' tab with the following output:

```
PS C:\Users\Umair_khan\Desktop\tutode-assignments> git --no-pager show 74212d1 --stat
>> git --no-pager show 7f011ce --stat
commit 74212d194193b81586631d85ded4984d2beed024
Author: Umair-bey <umair2025khan.ir@gmail.com>
Date:   Fri Oct 2 00:35:53 2026 +0530

    Task 4: Add Item ID field to To-Do form

Assignment-4/templates/todo.html | 15 ++++++-----
1 file changed, 9 insertions(+), 6 deletions(-)
commit 7f011ceb6d103086600d4cc4073f51429b2fe4cf
Author: Umair-bey <umair2025khan.ir@gmail.com>
Date:   Fri Oct 2 15:10:10 2026 +0530

    Task 4: Re-commit - state with only Item ID field in To-Do form

Assignment-4/templates/todo.html | 6 +++++
1 file changed, 6 insertions(+)
```

Shows difference between original Item ID commit (74212d1) and re-commit (7f011ce)

Step 5: Rebase master_1 onto Updated main

Command:

```
git checkout master_1
git rebase main
```

Conflict in todo.html — resolved by keeping all 5 fields
Result: Successfully rebased and updated refs/heads/master_1

Step 6: Force-Pushed master_1

Command:

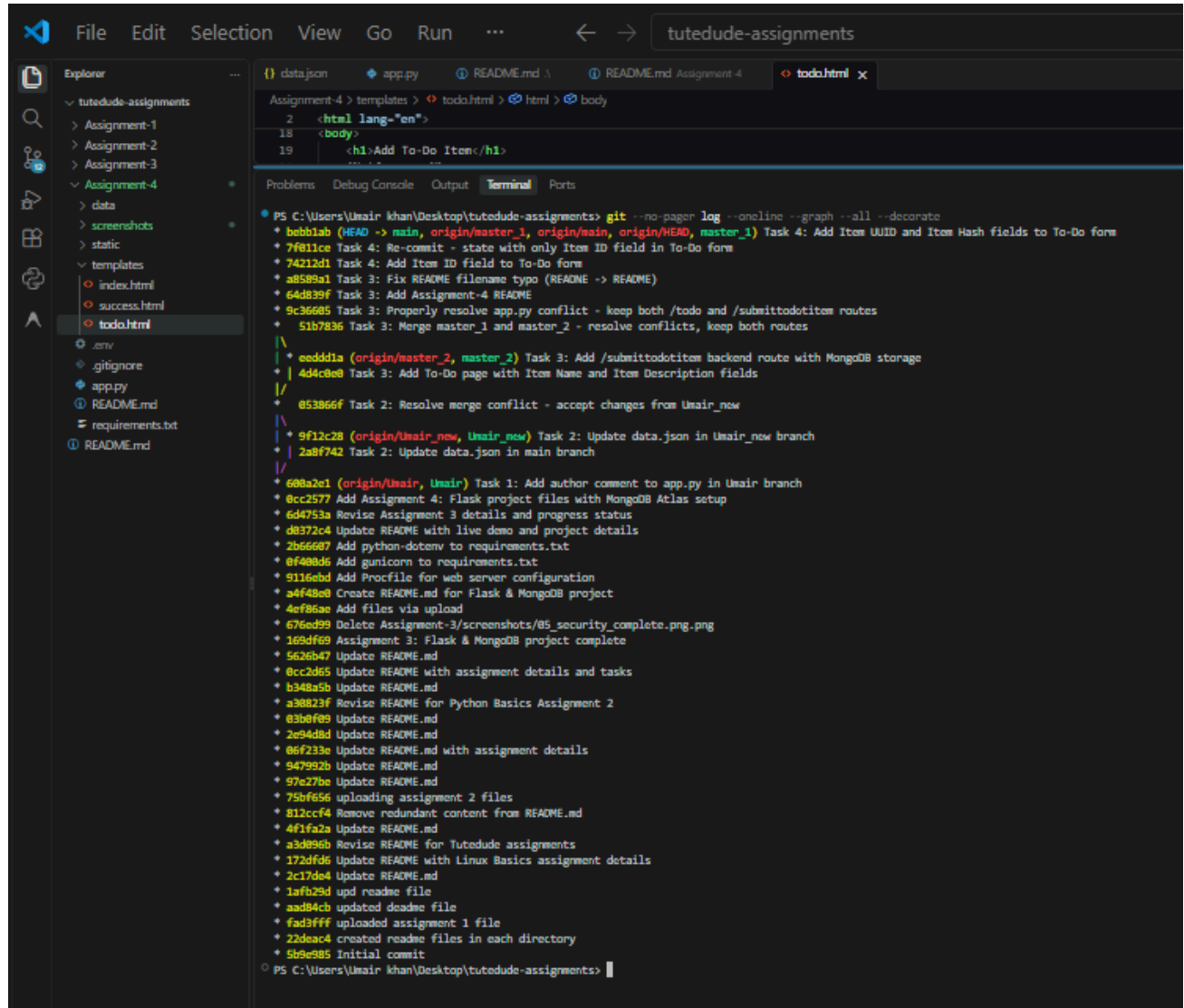
```
git push origin master_1 --force
```

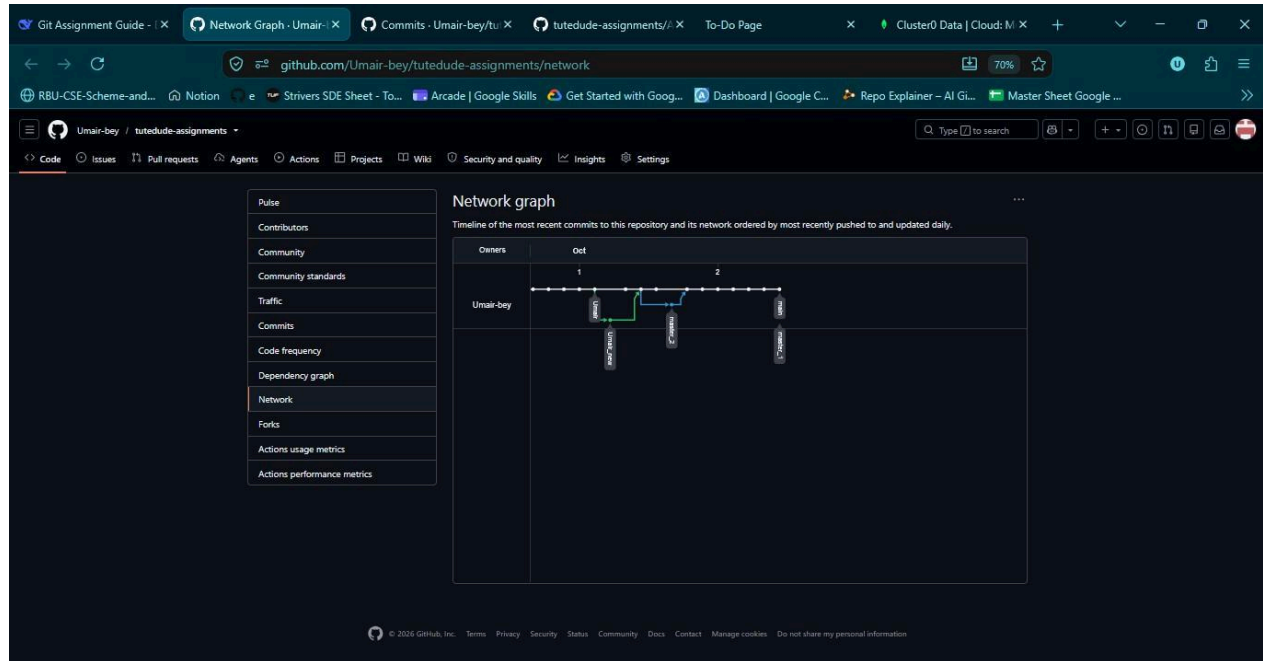
Step 7: Merged Final State into main

Commands:

```
git checkout main  
git merge master_1  
git push origin main
```

Step 8: Final Commit Graph





Confirms: sequential commits, reset, and rebase are all visible in history

=====

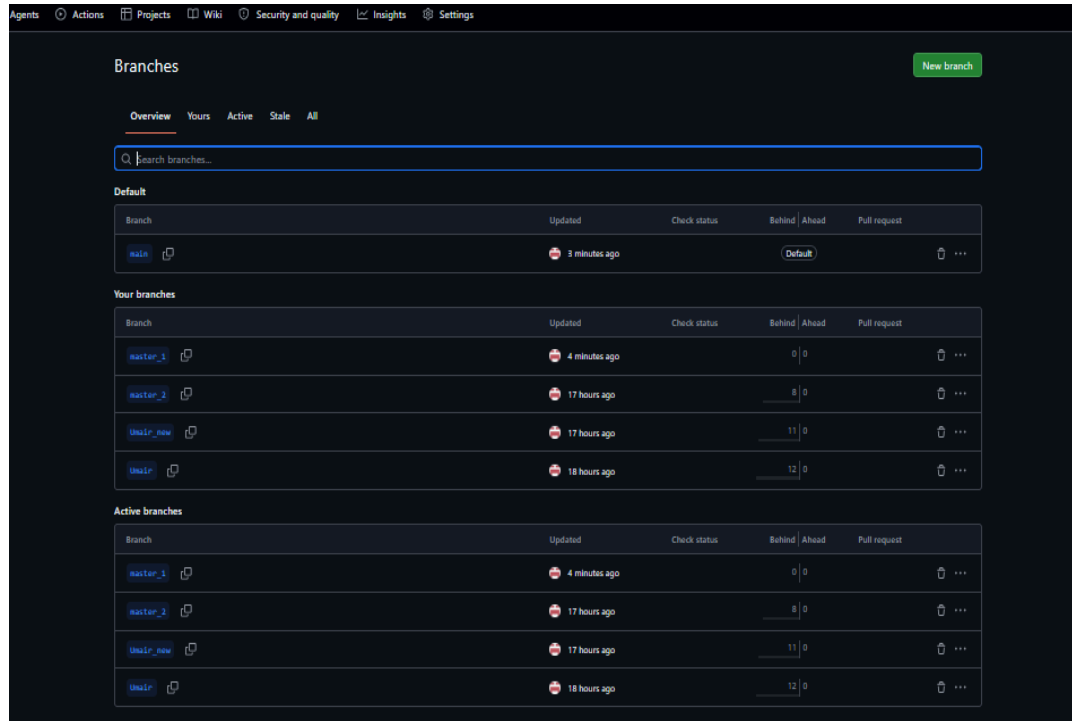
6. GITHUB REPOSITORY

=====

Repository: <https://github.com/Umais-bey/tutedude-assignments>

Assignment 4 folder:

<https://github.com/Umais-bey/tutedude-assignments/tree/main/Assignment-4>



=====

6.5 LIVE DEPLOYMENT

=====

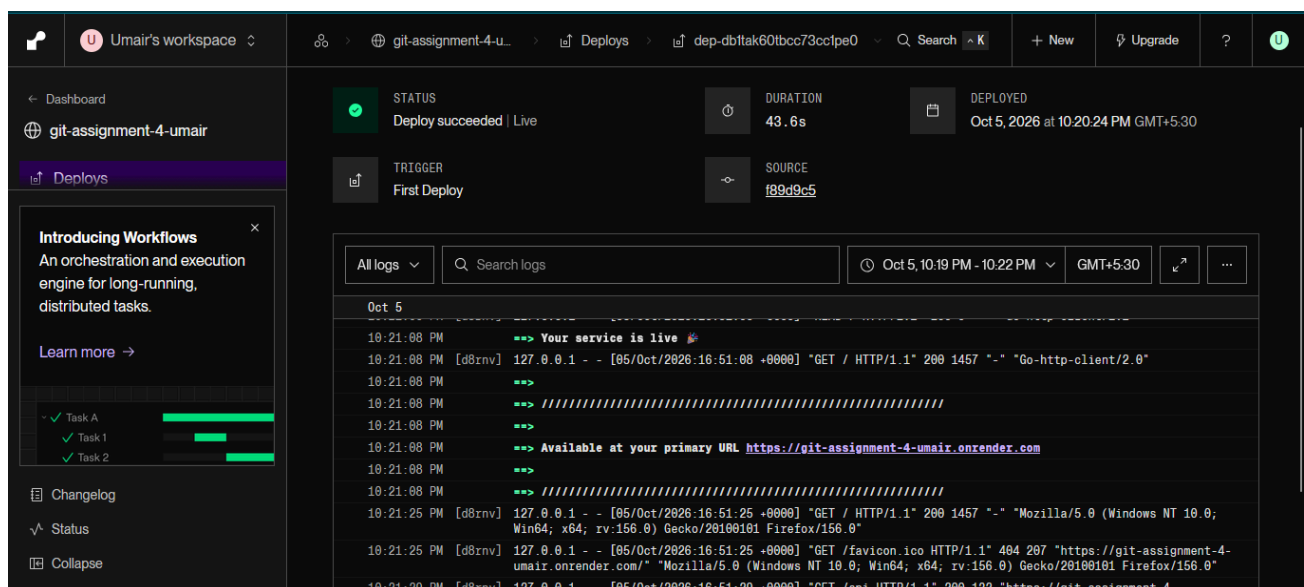
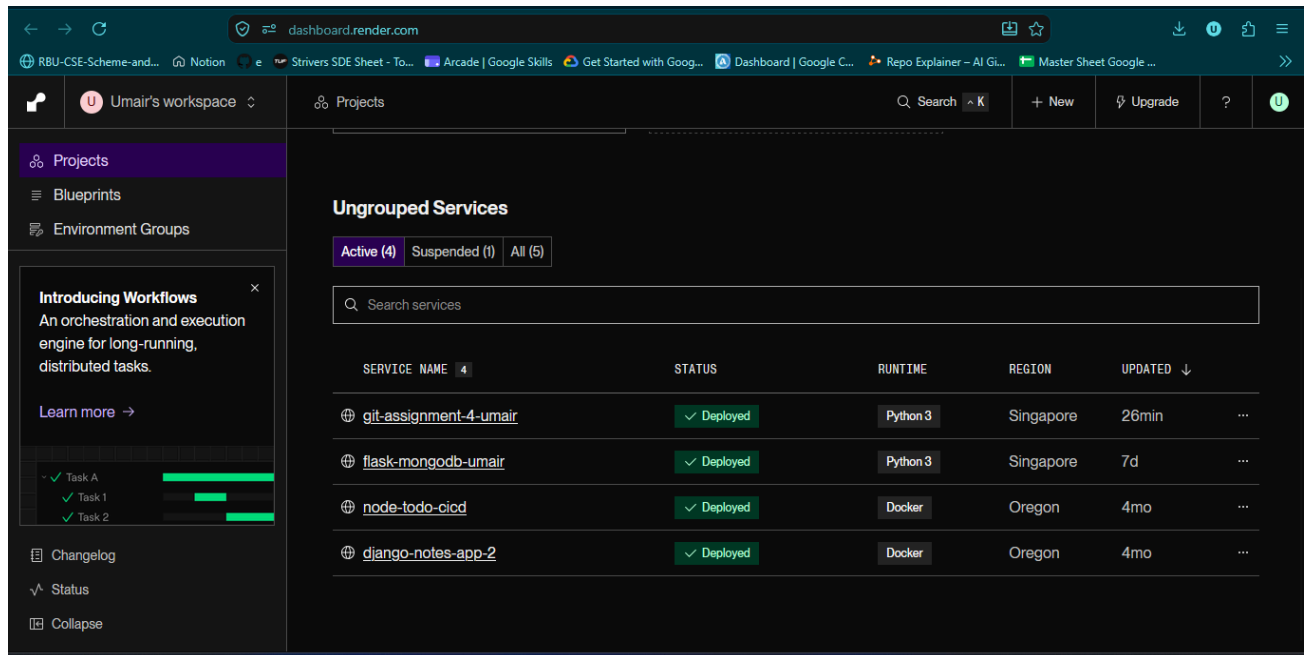
The Flask application was deployed to production using Render.com.

DEPLOYED URL:

<https://git-assignment-4-umair.onrender.com>

CONFIGURATION:

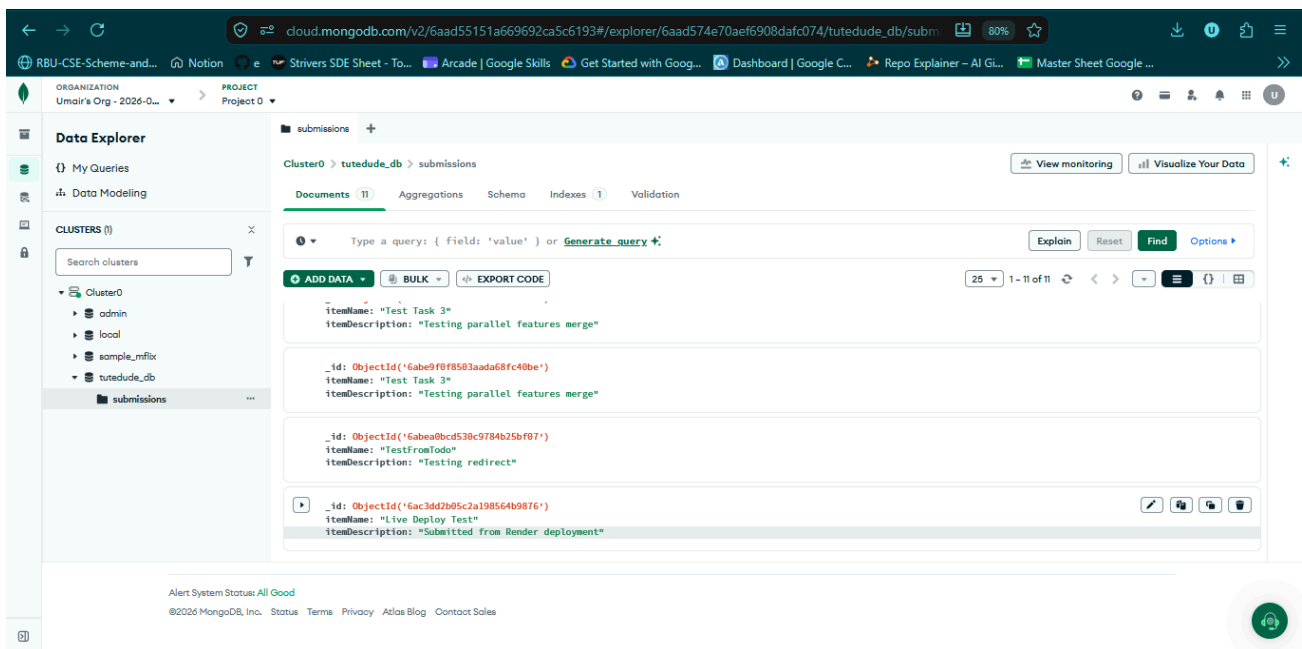
- Platform: Render.com (Free tier)
- Root Directory: Assignment-4
- Build Command: `pip install -r requirements.txt`
- Start Command: `gunicorn app:app`
- Environment Vars: MONGO_URI, FLASK_DEBUG=False



LIVE TESTING:

- ✓ Home page — loads successfully
- ✓ /todo — To-Do form with all 5 fields works
- ✓ /api — returns JSON data
- ✓ To-Do submissions save to MongoDB Atlas

The screenshot shows a web browser window with the URL `https://git-assignment-4-umair.onrender.com/todo`. The page has a dark theme and a central form titled "Add To-Do Item". The form contains five input fields: "Item ID:", "Item UUID:", "Item Hash:", "Item Name:", and "Item Description:". Each field has a placeholder text: "Enter item ID", "Enter item UUID", "Enter item hash", "Enter item name", and "Enter item description" respectively. Below the fields is a green button labeled "Add Item".



Note: Render's free tier sleeps after 15 min of inactivity.

First request after sleep takes 30–60 seconds.

7. CONCLUSION

All 4 tasks completed successfully:

Task 1: SSH setup, branch Umair created and merged to main

Task 2: Branch Umair_new, JSON conflict resolved by accepting _new changes

Task 3: Parallel branches master_1 (frontend) + master_2 (backend API),

merged with MongoDB Atlas integration

Task 4: Sequential commits (ID → UUID → Hash), soft reset, rebase,
and final merge

PROJECT STRUCTURE

Assignment-4/

```
|— app.py
|— requirements.txt
|— .env          (gitignored — contains MongoDB URI)
|— .gitignore
|— data/
|   |— data.json
```



```
|— templates/
|   |— index.html
|   |— success.html
|   |— todo.html
|— static/
|— screenshots/
|   |— [all task screenshots]
|— README.md
```

- GitHub Repository: <https://github.com/Umair-bey/tutedude-assignments>